

Chapter 1: Controlled Experiments

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method: **Comparison**

- ▶ Divide people (subjects) into two groups;
- ▶ Give the vaccine to one group (*treatment group*) and not to the other one (*control group*);
- ▶ Compare the responses (e.g. the percentage of subjects getting the disease) in each group.
 - ▶ If the percentage is lower in the treatment group, then the vaccine is effective; otherwise not effective.

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- Vaccine
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 - ▶ **Issue:** ?

Example: Polio Vaccine (First Experiment)

NFIP divided the subjects by school grade.

NFIP vaccinated all grade 2 children whose parents consented.

- ▶ Control: Grades 1 & 3;
- ▶ Treatment: Grade 2

	Grade	Size	Number Infected	Rate (per 100,000)
Treatment	2	225,000	56	25
Control	1 & 3	725,000	392	54
Not consent	2	125,000	55	44

Should we

- ▶ compare Treatment v.s Control?
 - ▶ why comparing rates, not counts?
 - ▶ other issue?
- ▶ compare Treatment (Consent) v.s. (Not Consent)?

Why Compare the **Rates**?

In the U.S., the number of deaths from all causes is

- ▶ 2.1 million in 1990;
- ▶ 1.7 million in 1960

nearly a 25% increase.

True or false, and why:

The data show that the public's health got worse over the period 1960-1990. (The population in 1960 was about 180 million, in 1990 about 248 million)

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The data show that the public's health got worse over the period 1960-1990. (The population in 1960 was about 180 million, in 1990 about 248 million)

False. The population got bigger too.

Death rate in 1960 = $1.7/180 \times 100\% = 0.94\%$

Death rate in 1990 = $2.1/248 \times 100\% = 0.85\%$

The death rate actually dropped over the period 1960-1990.

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Definition: Confounders/Confounding Variables

Factors that can also explain the difference in the outcomes of the treatment group and the control group

Polio Vaccine: Volunteers/Non-Volunteers

- ▶ Need parents' permission to vaccinate their kids
- ▶ If treatment = volunteers, control = non-volunteers,
 - ▶ Higher income parents would more likely consent to vaccinating their child;
 - ▶ Higher income → more hygienic environment → less chance to contract mild cases of polio early in childhood → less chance to develop antibodies to polio;
 - ▶ So, volunteers tend to be more vulnerable to polio;
 - ▶ The results will be **biased against** the new vaccine;
- ▶ The effect of being a volunteer/non-volunteer will be **confounded** with the effect of the treatment.
- ▶ Conclusion: To avoid **confounding**, *both* the treatment group and control group must be chosen from volunteers;

Possible Strategies to Deal with Confounders

For example, **age** is confounded with the effect of the vaccine:

- ▶ **restrict** the confounder (e.g. study only 6-yr-old kids)
- ▶ **control for** the confounder (e.g. conduct separate experiments for each age group)
- ▶ **balance** (e.g. make the age compositions in the treatment and control group similar)

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A simple way to ensure balance: *Randomization*, which means assigning subjects to the two groups *randomly*.

- ▶ Randomly $\neq$ Haphazardly
- ▶ Avoid the human factor: use coin tossing/random number generation.

Example: Polio Vaccine (Randomized Experiment)

- ▶ Of the children whose parents consented to vaccination, half were randomly assigned to the control group and the other half to the treatment group.
 - ▶ Hence this is a **randomized controlled** experiment.
- ▶ **Blinding**: The children in the control group were given a **placebo**.
 - ▶ Ensures response is to the vaccine, not the *idea* of vaccination.
- ▶ Doctors were unaware of which group each child was in.
 - ▶ This is known as **double blinding** since neither the child nor the doctor knows whether they got the vaccine or the placebo.

Blinding & Placebo

Blinding: Keeping subjects from knowing which group they are in: give a “fake” treatment (**placebo**) to the control group.

- ▶ Knowing if you are being treated or not might be confounding.
- ▶ Example: Pain killer

	Study 1 (w. placebo)	Study 2 (w/o. placebo)
Treatment	30%	31%
Control	31%	20%

Is the drug effective? Which study should we look at?

Single Blind v.s. Double Blind

Single Blind: Only the subject are kept from knowing which treatment they received.

Double Blind: Neither the subject nor those who evaluate the outcome (e.g., diagnosing the disease) know which treatment the subject receive

- ▶ To avoid the bias in the evaluation, especially in the borderline cases.

Example: Polio Vaccine Results

Randomized controlled double-blind experiment

	Size	Rate (per 100,000)
Treatment	200,000	28
Control	200,000	71
No consent	350,000	46

- Which pair of rates shows the vaccine is effective?
(a) 28 and 71 (b) 28 and 46 (c) 71 and 46
- Neither the control group nor the non-consent group got the vaccine, but the no-consent group had a lower rate of polio. Why?
- In a badly designed test where all volunteers are assigned to the treatment group, and all non-volunteers are to the control group, which pair of rates are the most likely rates for the two groups?
(a) 28 and 71 (b) 28 and 46 (c) 71 and 46

Comparison of the Two Polio Vaccine Experiments

Randomized controlled double-blind experiment			NFIP study		
	Size	Rate	Grade	Size	Rate
Treatment	200,000	28	2 (vaccine)	225,000	25
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- The two vaccine groups had similar rates of polio, 28 and 25. The two no-consent groups also had similar rates of polio, 46 and 44. Why are the rates of polio in the two control groups so different?
- In the NFIP study, neither the control group nor the no-consent group got the vaccine. Yet the no-consent group had a lower rate of polio. Why?
- Can one conclude that the vaccine is effective by comparing the 44 in the NFIP study with the 25 in the vaccine group?

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- Can one conclude that the vaccine is effective by comparing the 44 in the NFIP study with the 25 in the vaccine group?
 - No! volunteers/non-volunteers comparison

Historical and Contemporaneous Controls

- ▶ Sometimes randomized controlled experiments are difficult, expensive, or unethical to do.
- ▶ One commonly used (but poor) alternative is to compare patients using new treatment with **historical controls**: patients treated in the old way **in the past**.
- ▶ In contrast, in randomized controlled experiments, subjects in both groups are chosen from a population at the same time period (**contemporaneous controls**).
- ▶ Historical controls may not be reliable because the treatment group and the historical control group may differ in important ways.